

Math140C - HW #7

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2026.06.07

Q9

Fix some $x \in (a, b)$ and take an arbitrary sequence (x_n) converging to x . Clearly $f(t)\chi_{[a, x_n]}(t) \rightarrow f(t)\chi_{[a, x]}(t)$ for every $t \in [a, b] \setminus \{x\}$. Further, $|f_n| \leq |f|$ on $[a, b]$, hence the dominated convergence theorem gives

$$\begin{aligned} F(x) &= \int_a^b f\chi_{[a, x]} dt = \lim_{n \rightarrow \infty} \int_a^b f\chi_{[a, x_n]} dt \\ &= \lim_{n \rightarrow \infty} \int_a^{x_n} f dt \\ &= \lim_{n \rightarrow \infty} F(x_n) \end{aligned}$$

(x_n) is an arbitrary sequence converging to x , so F is continuous at x .

Q10

Recall $f \in \mathcal{L}^1(X, \mu) \iff |f| \in \mathcal{L}^1(X, \mu) \iff \int_X |f| d\mu < \infty$. Let $A = \{x \in X \mid |f(x)| < 1\}$ and $B = \{x \in X \mid |f(x)| \geq 1\}$ partition X .

$$\begin{aligned} \int_X |f| d\mu &= \int_A |f| d\mu + \int_B |f| d\mu \\ &\leq \int_A 1 d\mu + \int_B |f|^2 d\mu \\ &= m(A) + \|f\chi_B\|_2^2 \end{aligned} \tag{1}$$

The first term in (1) is finite because $A \subset E$ and $m(E) < \infty$ by assumption. The second term in (1) is also finite because $f \in \mathcal{L}^2(X, \mu)$ and $|f\chi_B| \leq |f|$ on E . This shows $f \in \mathcal{L}^1(X, \mu)$.

Next, let $f(x) = \frac{1}{1+|x|}$. Notice f is symmetric about $x = 0$. f is nonnegative everywhere and decreases monotonically on $[1, \infty)$, so one can apply the integral test to f^2 :

$$\sum_{n=1}^{\infty} f(n)^2 \text{ converges} \iff \int_1^{\infty} f^2 dx \text{ converges} \tag{2}$$

$\sum_{n=1}^{\infty} f(n)^2 = \sum_{n=1}^{\infty} \frac{1}{(1+n)^2}$ converges, so the integral in (2) converges and is finite. By symmetry, $\int_{-\infty}^{-1} f^2 dx$ converges and is finite. Of course, $\int_{-1}^1 f^2 dx$ is finite, hence

$$\int_{\mathbb{R}} |f|^2 dx < \infty,$$

Meanwhile, a similar analysis via the integral test applied to f shows

$$\int_{\mathbb{R}} f \, dx = \infty$$

because $\sum_{n=1}^{\infty} f(n) = \sum_{n=1}^{\infty} \frac{1}{1+n}$ diverges. The conclusion is that $f \in \mathcal{L}^2(\mathbb{R}, m)$ but $f \notin \mathcal{L}^1(\mathbb{R}, m)$.

Q12

Yes, g is continuous on $[0, 1]$. To emphasize which variable is fixed, let

$$\begin{aligned} f^x(t) &= f(x, t) \text{ be a function in } y \\ f^y(s) &= f(s, y) \text{ be a function in } x. \end{aligned}$$

Pick an arbitrary $x \in [0, 1]$ and take any sequence (x_n) converging to x . Since $f^y(s)$ is continuous for any fixed s ,

$$f^{x_n}(t) \rightarrow f^x(t) \text{ upon fixing any } t \in [0, 1].$$

By assumption, $|f^{x_n}(t)| \leq 1$ for all $t \in [0, 1]$, so dominated convergence theorem gives

$$\begin{aligned} g(x) &= \int_0^1 f^x(y) \, dy = \lim_{n \rightarrow \infty} \int_0^1 f^{x_n}(y) \, dy \\ &= \lim_{n \rightarrow \infty} g(x_n) \end{aligned}$$

(x_n) is an arbitrary sequence converging to x , so g is continuous at x .

Q16

Note E is measurable by Question 3 of Rudin. Take any measurable $A \subset E$. Clearly, $\chi_A \in \mathcal{L}^2([-\pi, \pi], m)$ and, with respect to (e^{inx}) , has Fourier coefficients

$$\begin{aligned} c_n &:= \frac{1}{2\pi} \int_{-\pi}^{\pi} \chi_A e^{-inx} \, dx \\ &= \frac{1}{2\pi} \int_A (\cos nx - i \sin nx) \, dx \end{aligned}$$

$\sum_{n=-\infty}^{\infty} |c_n|^2$ converges, so (c_n) and all its subsequences converge to 0. Namely, $c_{n_k}, c_{2n_k} \rightarrow 0$ as $k \rightarrow \infty$. This gives

$$\int_A \sin n_k x \, dx \rightarrow 0$$

and

$$\begin{aligned} 2 \int_A (\sin n_k x)^2 \, dx &\stackrel{\text{trig}}{=} \int_A (1 - \cos 2n_k x) \, dx \\ &\rightarrow \int_A 1 \, dx = m(A) \end{aligned}$$

as $k \rightarrow \infty$, proving the two hints given by Rudin.

Define

$$f(x) := \lim_{k \rightarrow \infty} \sin n_k x, \quad (x \in E).$$

Take any measurable $A \subset E$. On A , the sequence $(\sin n_k x)_k$ still converges to $f(x)$ pointwise; further, the sequence is dominated by 1, so the results from the hint give

$$\int_A f dm = 0 \tag{3}$$

$$\int_A f^2 dm = \frac{1}{2} m(A) \tag{4}$$

(4) can be rearranged as $\int_A (f^2 - \frac{1}{2}) dm = 0$; Letting $A = E$, the result from Question 2 of Rudin says

$$f = \pm \sqrt{\frac{1}{2}} \text{ almost everywhere on } E \tag{5}$$

Let $A_1 = \{x \in E \mid f(x) = +\sqrt{\frac{1}{2}}\}$ and $A_2 = \{x \in E \mid f(x) = -\sqrt{\frac{1}{2}}\}$. Then

$$\int_{A_1} f dm = \sqrt{\frac{1}{2}} m(A_1), \quad \int_{A_2} f dm = -\sqrt{\frac{1}{2}} m(A_2),$$

and considering the cases $A = A_1$ and $A = A_2$ gives $m(A_1) = m(A_2) = 0$. Combining this result with (5) gives $m(E) = 0$, as was to be shown.

Q17

Once again, clearly $\chi_E \in \mathcal{L}^2([-\pi, \pi], m)$. With respect to the orthogonal system (e^{inx}) , it has Fourier coefficients

$$\begin{aligned} c_n &:= \frac{1}{2\pi} \int_{-\pi}^{\pi} \chi_E e^{-inx} dx \\ &= \frac{1}{2\pi} \int_E e^{-inx} dx \\ &= \frac{1}{2\pi} \int_E (\cos nx - i \sin nx) dx \end{aligned}$$

For contradiction, suppose there exists $\delta > 0$ such that infinitely many integers n satisfy

$$\forall x \in E, \quad \sin nx > \delta$$

Then for such integers n ,

$$\begin{aligned} 4\pi^2 |c_n|^2 &= \left(\int_E \cos nx dx \right)^2 + \left(\int_E \sin nx dx \right)^2 \\ &\geq \delta^2 m(E)^2 \end{aligned}$$

That there are infinitely many such n contradicts a consequence of Bessel's inequality, $c_n \rightarrow 0$.